

COURSE: CHE 312: Transport Phenomena II

Fall Semester 2025

Department of Chemical Engineering

University of Illinois Chicago

Exam #3

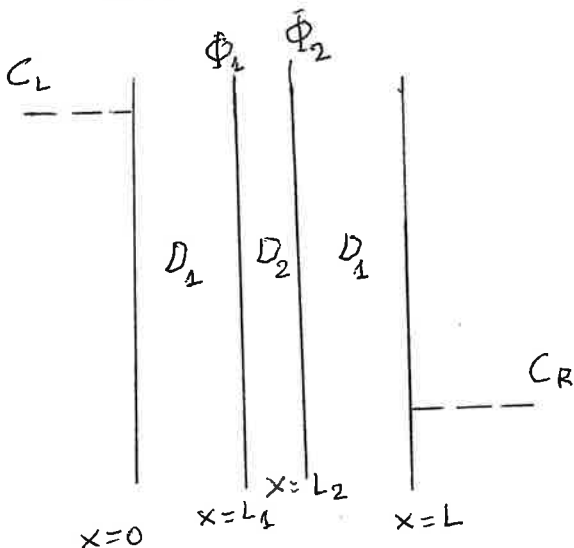
Duration: 70 minutes --- November 25, 2025

All Problems must be done

Problem 1: (30 points)

Consider steady state diffusion of the species A through three media arranged as shown in the attached diagram. Assume that diffusion is one-dimensional. At $x = 0$, $C_{A1} = C_L$, and at $x = L$, $C_{A3} = C_R$. In medium 2, there is a zeroth order chemical reaction through which the species A is consumed. Develop an expression for the concentration profiles across the three media.

(Bonus 10 points) Develop an expression for the steady state flux across the three media.



$$C_A(x=0) = C_L$$

$$C_A(x=L) = C_R$$

$$\frac{d^2 C_A}{dx^2} = 0$$

$$C_{A1}(x) = A_1 x + B_1$$

$$C_{A1}(0) = C_L = B_1$$

$$D_2 \frac{d^2 C_{A2}}{dx^2} = -\Gamma$$

$$\frac{dC_{A2}}{dx^2} = -\frac{\Gamma}{D_2}$$

$$\frac{dC_{A2}}{dx} = -\frac{\Gamma}{D_2} x + A_2$$

$$C_{A2}(x) = -\frac{\Gamma}{2D_2} x^2 + A_2 x + B_2$$

$$\frac{d^2 C_{A3}}{dx^2} = 0$$

$$C_{A3}(x) = A_3 x + B_3$$

$$C_{A3}(L) = A_3 L + B_3 = C_R$$

$$\phi = -D \frac{dC_A}{dx} = -D A$$

Problem 2: (70 points)

Consider the reversible conversion of species A into B through the stoichiometric reactions $A \xrightleftharpoons[k_{-1}]{k_1} B$ in a stationary liquid film of thickness L. Determine C_A and C_B throughout the film, if

(35 Points) the surface at $y = L$ is assumed to be impermeable and unreactive, and the concentrations of A and B at $y = 0$ are specified as C_{A0} and C_{B0} . The forward and reverse homogeneous reactions follow zeroth order kinetics

(35 Points) the surface at $y = L$ is assumed to be impermeable and unreactive, and the concentrations of A and B at $y = 0$ are specified as C_{A0} and C_{B0} . The forward and reverse homogeneous reactions follow zeroth and first order kinetics, respectively

$$D_A \frac{d^2 C_A}{dy^2} = -k_1 + k_{-1}$$

$$D_B \frac{d^2 C_B}{dy^2} = k_1 - k_{-1}$$

$$\frac{d^2 C_A}{dy^2} = \frac{-k_1 + k_{-1}}{D_A}$$

$$\frac{dC_A}{dy} = \frac{-k_1 + k_{-1}}{D_A} y + A$$

$$C_A = \frac{-k_1 + k_{-1}}{2D_A} y^2 + Ay + B$$

$$C_B = \frac{k_1 - k_{-1}}{2D_B} y^2 + Cy + D$$

